

Nexalytics HR Solutions

Issued: May 04, 2026 | Priority: **HIGH**
| Deadline: 1 Week

1. COMPANY OVERVIEW

Nexalytics HR Solutions is a mid-sized technology and professional services firm operating across four regional offices in the United States. The People Analytics team manages workforce data for 1,470 employees across three departments — Sales, Research & Development, and Human Resources. Despite competitive compensation, voluntary employee turnover has reached **16.1%** — significantly above the industry benchmark of 10%. Leadership has commissioned this analytics project to identify root causes and recommend data-driven retention strategies.

2. BUSINESS PROBLEM

The HR Director needs data-driven answers to:

- Which employees are most likely to leave — and why?
- Which departments and job roles have the highest attrition rates?
- Does overtime, salary, or job satisfaction predict attrition most strongly?
- What is the financial cost of replacing employees who leave?
- Which employee segments need immediate intervention?
- Where should HR focus retention budgets for maximum impact?

3. DATASET OVERVIEW — WA_FN-USEC_-HR-EMPLOYEE-ATTRITION.CSV

Company	Nexalytics HR Solutions	Total Records	1,470 employees
Source	IBM HR Analytics Dataset	Target Variable	Attrition (Yes / No)
Columns	35 features	Attrition Rate	16.1% (237 leavers)

Column	Type	Description
Age	Integer	Employee age in years
Attrition	Text	Target: Yes = left company, No = stayed
BusinessTravel	Text	Non-Travel / Travel_Rarely / Travel_Frequently
Department	Text	Sales / Research & Development / Human Resources
DistanceFromHome	Integer	Miles from home to office
Education	Integer	1 (Below College) → 5 (Doctor)
EnvironmentSatisfaction	Integer	1 (Low) → 4 (Very High)
JobInvolvement	Integer	1 (Low) → 4 (Very High)
JobLevel	Integer	1 (Junior) → 5 (Senior Executive)
JobRole	Text	9 roles (Sales Exec, Research Scientist, etc.)

JobSatisfaction	Integer	1 (Low) → 4 (Very High)
MonthlyIncome	Integer	Monthly salary in USD
NumCompaniesWorked	Integer	Total prior employers
OverTime	Text	Yes / No — key attrition driver
PercentSalaryHike	Integer	Last appraisal raise percentage
RelationshipSatisfaction	Integer	1 (Low) → 4 (Very High)
StockOptionLevel	Integer	0–3 stock option grant level
TotalWorkingYears	Integer	Total years in the workforce
WorkLifeBalance	Integer	1 (Bad) → 4 (Best)
YearsAtCompany	Integer	Tenure at current company
YearsInCurrentRole	Integer	Years in current job role
YearsSinceLastPromotion	Integer	Years since last promotion

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DATA QUALITY & BUSINESS RULES

Issue	Details	Action Required
Useless columns	EmployeeCount (all=1), Over18 (all=Y), StandardHours (all=80)	Drop all three — zero variance
No NULL values	Dataset is complete — no missing fields	No imputation needed
Class imbalance	84% No / 16% Yes — skewed target	Use SMOTE or class weights in model
Satisfaction scale	All satisfaction scores rated 1–4	Treat as ordinal, not continuous
Salary outliers	MonthlyIncome ranges \$1,009 – \$19,999	Check for executive-level skew

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KPIS TO BUILD

#	KPI Name	Formula	Page
1	Overall Attrition Rate	Leavers / Total Employees × 100	1
2	Total Employees	COUNT(EmployeeNumber)	1
3	Avg Monthly Income	AVG(MonthlyIncome)	1
4	Overtime Attrition Rate	Leavers with OT / Total OT Employees × 100	1
5	Dept Attrition Rate	Leavers per Dept / Dept Headcount × 100	2
6	Role Attrition Rate	Leavers per JobRole / Role Headcount × 100	2
7	Income vs Attrition	AVG(MonthlyIncome) by Attrition bucket	2
8	Satisfaction vs Attrition	AVG(JobSatisfaction) by Attrition bucket	2
9	Tenure vs Attrition	AVG(YearsAtCompany) WHERE Attrition = Yes	3
10	Promotion Gap	AVG(YearsSinceLastPromotion) by Attrition	3

11	Distance Impact	AVG(DistanceFromHome) by Attrition bucket	3
12	Flight Risk Score	ML model probability score per employee	3

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. DASHBOARD BLUEPRINT (3 PAGES)

- KPI Cards: Total Employees, Attrition Rate, Avg Salary, OT Attrition %
- Monthly / Yearly Attrition Trend — Line Chart
- Attrition by Department — Bar Chart
- Attrition by Gender & Marital Status — Donut Charts
- Department and Year slicers

- Top 5 Job Roles by Attrition Rate — Bar Chart
- Salary Band vs Attrition — Clustered Bar Chart
- Overtime vs Non-Overtime Attrition — Comparison Bar
- Job Satisfaction vs Attrition — Scatter / Box Plot
- JobLevel and Department slicers

- Flight Risk Employee Table — sortable by risk score
- Distance from Home vs Attrition — Bar Chart
- Years at Company vs Attrition Heatmap
- Promotion Gap Analysis — Bar Chart
- Top 5 Attrition Drivers — Feature Importance Chart
- Warehouse and Role slicers

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. DELIVERABLES & TIMELINE

Day	Task	Deliverable	Status
1	Setup + Excel Exploration	Data Dictionary + EDA Notes	Pending
2	Python EDA + Cleaning	cleaned_hr_attrition.csv + .ipynb	Pending
3	Python Advanced Analysis	KPI outputs + attrition charts saved	Pending
4	SQL Queries	15+ SQL KPI queries .sql file	Pending
5	Power BI Data Model	Star schema + relationships	Pending
6	Power BI Dashboard Build	3-page dashboard .pbix	Pending
7	GitHub + README	Public repo with all files + screenshots	Pending

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. MANAGER EXPECTATIONS

- Tell a story with data — every chart must answer a specific business question.
- Every insight needs a recommendation — don't just show the problem, suggest the fix.
- Cross-validate all numbers — Python totals must match SQL totals must match Power BI totals.

- Clean, commented code — SQL scripts and Python notebooks must be readable by anyone.
- Professional GitHub — README must explain the project clearly with screenshots.
- Quantify the impact — always express findings in dollar or percentage terms.
- Root cause first — do not stop at 'Sales has high attrition'; explain WHY.

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HIDDEN INSIGHTS IN THE DATA (FIND THESE!)

#	Finding	Expected Impact
1	Sales dept accounts for 38%+ of all attrition	Department-level retention program
2	Overtime employees leave at 3× the rate of non-OT staff	Overtime policy overhaul
3	Employees aged 25–35 have highest attrition	Early-career engagement strategy
4	Low job satisfaction (score 1) predicts 50%+ attrition	Manager effectiveness review
5	Single employees leave at 2× rate of married employees	Benefits & culture redesign
6	No promotion in 4+ years strongly predicts leaving	Promotion cadence policy
7	Monthly income below \$3,500 doubles attrition risk	Salary band benchmarking
8	Laboratory Technicians have the highest role-level attrition	Targeted retention bonus